

Hydrostatic shape of Earth

Hydrostatic stress: $\underline{\underline{\sigma}} = -p \underline{\underline{I}}$ $p = p_0 - \rho |g| x_3$

Substitute into eqbm: $\nabla \cdot \underline{\underline{\sigma}} + \rho \underline{g} = 0$

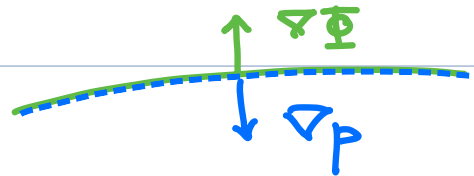
$$\nabla \cdot \underline{\underline{\sigma}} = \nabla \cdot (-p \underline{\underline{I}}) = -(\underline{\underline{I}} \nabla p + p \nabla \cdot \underline{\underline{I}}) = -\nabla p$$

$$\Rightarrow \nabla p = \rho \underline{g} \quad \nabla p \parallel \underline{g}$$

Grav. field: $\underline{g} = -\nabla \Phi$ Φ grav. potential

so that

$$\boxed{\nabla p = -\rho \nabla \Phi}$$

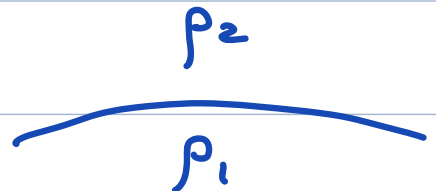


$\Rightarrow$ Isobars and equipotentials are parallel

isobar: $p(\underline{x}) = p_1$

equipotential: $\Phi(\underline{x}) = \Phi_1$

Surface of planet is change
in density of fluid



$\Rightarrow$ isopycnic

$$\begin{aligned} \underline{0} &= \nabla \times \nabla p = \nabla \times (\rho \underline{g}) = \nabla \rho \times \underline{g} + \rho \nabla \times \underline{g} \\ &= -\nabla \rho \times \nabla \Phi - \rho \nabla \times \nabla \Phi \end{aligned}$$

$$\nabla \rho \times \nabla \Phi = 0 \Rightarrow \nabla \rho \parallel \nabla \Phi$$

$\Rightarrow$ surface of body in hydrostatic equilibrium
is an equipotential surface

I) Stationary body

1) Grav. field eqn: $\underline{g} = -\nabla \Phi$

2) Gauss' law: $\nabla \cdot \underline{g} = -4\pi G \rho$

$$-\nabla \cdot \nabla \Phi = -4\pi G \rho$$

$$\boxed{\nabla^2 \Phi = 4\pi G \rho}$$

In spherical coordinates: $\nabla = \frac{\partial}{\partial r}$

(not a cartesian frame) $\nabla \cdot = \frac{1}{r^2} \frac{\partial}{\partial r} r^2$

For a homogeneous spherical body for radius R :

$$\boxed{\Phi = \begin{cases} \frac{4}{6} \pi G \rho (r^2 - R^2), & r \leq R \\ -3 \pi G \left(\frac{1}{r} - \frac{1}{R} \right), & r > R \end{cases}}$$

$\Rightarrow \Phi = \Phi(r)$ equipotentials are spheres

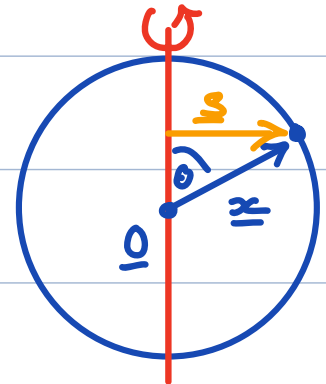
II, Rotating body

In frame rotating with body need to add centrifugal force $\underline{f}_c = m \Omega^2 \underline{s}$

Ω = angular velocity

$\underline{s}$ = distance from axis

$$\underline{s} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \quad s = |\underline{s}|$$



$$\theta = \text{polar angle} \quad |\underline{s}| = |\underline{x}| \sin \theta$$

adds a "fictitious" acceleration & potential

centrifugal: $\underline{g}_c = \Omega^2 \underline{s}$

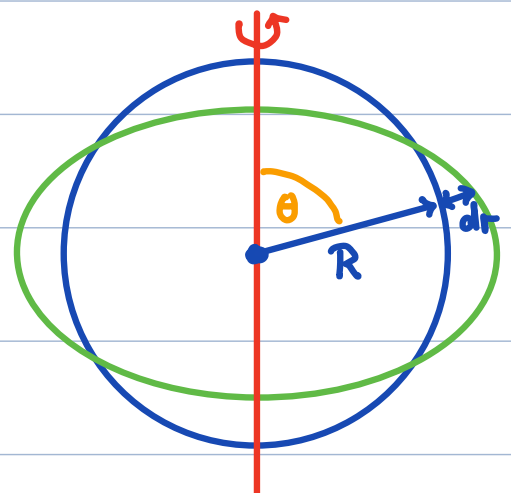
$$\Phi_c = -\frac{1}{2} \Omega^2 s^2$$

gravitational: $\underline{g}_G = \frac{GM}{R^2}$

$$\Phi_G \propto \int g \, dr = \frac{MG}{R^2} dr$$

Note Φ_G is linearized at $r=R$

and $dr(\theta)$ is deviation from sphericity



Total potential:

$$\Phi = \Phi_G + \Phi_c + \Delta\Phi$$

$\Delta\Phi$ = self-potential
due to deformation

ignoring $\Delta\Phi$

$$\Phi = \frac{MG}{R^2} dr(\theta) - \frac{\Omega^2 R^2}{2} \sin^2 \theta$$

solving for dr

$$dr = \frac{R}{2} q \sin^2 \theta + \text{const}$$

$$q = \frac{\Omega^2 R}{|g|} = \frac{\Omega^2 R^3}{MG}$$

Constant from mass/volume conservation

$$\int_0^\pi dr(\theta) dS = 2\pi R^2 \int_0^\pi dr(\theta) \sin \theta d\theta = 0$$

$$\Rightarrow \boxed{dr = dr_0 \left(\sin^2 \theta - \frac{2}{3} \right)} \quad dr_0 = \frac{1}{2} R q$$

$$\text{Earth: } R = 6371 \text{ km} \quad |g| = 9.81 \frac{\text{m}}{\text{s}^2} \quad \Omega = 76 \cdot 10^{-6} \frac{1}{\text{s}}$$

$$\Rightarrow q \approx 3.5 \cdot 10^{-3} \quad \Rightarrow dr_0 \approx 11 \text{ km}$$

$$\text{actual } dr_0 = 21.4 \text{ km}$$

Error is due to self-potential

$$\Delta \bar{\Phi} = - \frac{3}{5} \frac{\rho_1}{\rho_0} g_s dr(\theta)$$

where $\rho_0 = \frac{M}{V}$ and $\rho_1 = \text{density of shifted material}$

$$\Rightarrow dr_0 = \frac{1}{2} R q \left(1 - \frac{3}{5} \frac{\rho_1}{\rho_0} \right)^{-1} \approx 21.9 \text{ km}$$

$$\text{for } \rho_0 = 5500 \frac{\text{kg}}{\text{m}^3} \quad \rho_1 = 4500 \frac{\text{kg}}{\text{m}^3}$$